

Soft mediation, scalar-loop closure and protected flavor

A stable 48-scalar completion, exact CP operator census and representation-enforced coefficient relations

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Complete mediator and fixed-mass calculations with explicit Gaussian inputs. Scalar-loop closure and fermion protection are established within their stated scopes; golden flavor is not predicted.

Central results

The source coefficients now have a constructive renormalizable mediator realization, and their scalar-loop closure is executable. Three finite-group mediator multiplets add 27 real coordinates to the existing 21-coordinate Higgs/nonet action. The resulting 48-scalar vacuum has a positive full Hessian, exactly the same source-frame CKM observables and finite quark masses, and the existing termwise first-order one-loop mass-phase protection. Their representations forbid new renormalizable quark bilinears, even when every permitted scalar interaction is admitted. This is a structural protection result for a class of scalar completions, rather than another fitted angular potential.

Exact character and generalized-CP coset arithmetic gives 251 permitted scalar coefficients on this field content. The constructed Gaussian benchmark chooses many of these coefficients to vanish; it does not suppress them by miscounting the operator space. Power counting identifies a radiatively closed soft-mediator restriction on the hard mediator couplings. Finite matching and low-energy running still generate calculable interactions. Six independent current coefficients move all four mixing magnitudes at fixed complete quark spectra, so the mediator realization does not enforce golden CKM depth or a 66-degree standard phase.

The scalar-loop compiler reconstructs the commutative, nonassociative one-loop multiplication of all 52 source quartics, with independent off-grid polynomial checks. Two radial quartics and just one finite-eight cross interaction generate the complete nine-operator adjoint-pair subalgebra. Combining the three finite channels into an SU(3) 27-plet instead gives a five-operator closed subalgebra and an enforced equality of their coefficients. That continuous-symmetry quartic limit cannot isolate an interior CP-breaking relative frame at fixed spectra. These results locate a concrete interaction boundary between protected coefficient relations and unconstrained flavor.

Tree matching and the minimum mediator content

The kinetic action assigns one half the squared derivative to each eta, each s and every orthonormal adjoint coordinate, together with canonical mediator coordinates and the usual Higgs kinetic term. This fixes the independent singlet normalization; $H=s I+A$ is the nonet coordinate convention. Source kinetic metrics and their flavor couplings are declared inputs. The scalar-only one-loop quartic calculation uses this same canonical metric.

Let q contain the four sector-quadratic singlets eta squared, s squared, eta s and the adjoint norm. Let b contain the three quadratic adjoint covariants eta A, s A and the traceless part of A squared. The remaining quadratic currents are the finite eight-prime, nine and CP-paired five projections of the symmetric-square adjoint vector w. The five types of current exhaust the possible linear mediator couplings to two source fields.

For a positive Gaussian mediator mass, eliminating the mediator gives minus one half the squared source current divided by mass squared. A shared singlet has current $g_u \dot{q}_u + g_d \dot{q}_d$. Its four-by-four mixed coefficient matrix is minus $g_u g_d$ -transpose divided by mass squared, so all two-by-two minors vanish. A shared ordinary adjoint gives the same rank-one structure for its three-by-three multiplicity matrix. A unique representation contraction leaves four singlet-current labels or three adjoint-current labels with independent real couplings; uniqueness does not select those multiplicity vectors.

One copy of each channel requires 36 real mediator coordinates and twenty cubic current coefficients. The mixed coefficient matrices have dimensions seven and five on their rank-one varieties; the three remaining scalar channels add three. Thus this tree restriction reduces the

28-dimensional mixed space to a generically fifteen-dimensional variety, of codimension thirteen. This dimension count is an exact algebraic statement on the specified Gaussian matching family. Direct source contacts and additional mediator interactions remain symmetry allowed.

Conversely, an arbitrary mixed matrix of rank r needs at least r copies of its mediator irrep. A singular-value decomposition constructs a positive-mass realization attaining this bound, including either sign of a cross coupling. Generic full-rank singlet and ordinary-adjoint matrices require four singlets and three ordinary adjoints. Together with one copy of each finite channel, the generic minimum is ten multiplets, or 55 real mediator coordinates. This universal construction reproduces all 28 mixed coefficients; it does not constrain them.

A stable 48-scalar completion with protected fermion vertices

Use only the finite eight-prime, nine and CP-paired five mediators. Their dimensions are eight, nine and ten. For a source current J and a positive mass M , write the UV scalar potential as the target source potential plus M squared times the norm of $S+J/M$ squared, divided by two. Expanding gives a source contact correction, a cubic $S \cdot J$ and a positive Gaussian mediator mass. The contact correction cancels precisely the three mixed finite-source quartics, transferring them to mediator exchange. The other source operators remain explicit inputs.

The mediator stationary value is minus J/M squared. At that value the source vacuum and all its observables are exactly those of the original source potential. The full Hessian has source block $H+D$ -transpose D/M squared, mixed block D and mediator block M squared I , where D is the current Jacobian. Its Schur complement is exactly H . Consequently every positive source Hessian has a positive full completion. The source potential's global lower bound and the added nonnegative square also establish boundedness; this does not certify the chosen local source minimum as the global vacuum.

The benchmark has 48 physical neutral real scalar coordinates and smallest Hessian eigenvalue approximately 1.126 times ten to the minus five. It retains J approximately 0.00026225346 and the same six light and six heavy quark masses. The current factors in the two sectors have different specified ratios, avoiding an accidental rank loss in a balanced factorization. Their values are declared inputs, without using a target angle or magnitude.

The quark triplet bilinear decomposes into a singlet and the ordinary adjoint eight. The finite eight-prime, nine and five-pair cannot appear in a renormalizable heavy-mass or heavy-to-light quark vertex. Thus the complete allowed renormalizable fermion action is unchanged by these 27 new scalar coordinates. All scalar mass eigenmodes may mix arbitrarily, but each quark source variation remains Hermitian. The paired mass-basis vertices are real term by term, preserving the first-order one-loop mass-phase result. This representation argument applies beyond the Gaussian potential to arbitrary allowed scalar coefficients, provided the vacuum and spectrum are defined. It does not extend the theorem to two loops or higher-dimensional fermion operators.

Complete scalar operator census and soft closure

The mediator representation W is Symmetric square of eight minus the singlet and ordinary eight, of real dimension 27. Before CP the dimensions of W quadratic, cubic and quartic invariant spaces are 4, 17 and 99. Generalized CP reduces them to 3, 11 and 58. For each source sector the source-squared W and source-squared W -squared dimensions are 4 and 77 before CP, and 3 and 50 after CP. With the previously complete source basis and Higgs portals, the total is 251 real scalar coefficients. W has no singlet, so no W tadpole or Higgs-squared linear- W coupling is available.

Every count, including the CP reduction, is computed exactly in $Q(\sqrt{5})$, ω . The method averages over the 1,080 family elements and over their antiunitary CP coset. On the Hermitian adjoint, an odd power of an antiunitary-induced real map has trace $\text{Tr}(T$ to that power) minus one, while an even power has trace equal to the squared modulus of $\text{Tr}(T$ to half that power), minus one. Here T is the unitary product $U U^\dagger$, with $U=g X$. The exact CP matrix's common normalization has squared value $(3+\sqrt{5})/128$, permitting the entire coset calculation without numerical trace rounding.

The Gaussian benchmark sets mediator self cubics, self quartics and quadratic-mediator/source-quadratic quartics to zero. They are allowed by the finite symmetry. Nevertheless, if W enters only through its positive mass and dimension-one source-current couplings, local ultraviolet power counting prevents divergent hard W quartic couplings from appearing. Every external W requires at least one dimension-one coupling; a divergent counterterm of dimension d has coefficient dimension $4-d$ and contains no inverse mass factors. Pure W cubics, pure W quartics and W -squared/source-squared quartics cannot meet that condition. This restriction is a perturbative soft-current property, not an assertion that finite effective interactions vanish.

An exact two-source fixture checks the distinction directly. Its separated source quartics plus a Gaussian singlet current have no hard mediator quartic or source cross-quartic one-loop counterterm. After eliminating the mediator, the low-energy exchange quartic does run into additional light interactions. A complete UV prediction must include its matching and running; failure of a tree coefficient identity in the EFT does not by itself destroy UV predictivity. Source contact quartics still require their complete allowed counterterms, and additional hard mediator couplings would remove the soft restriction.

Actual UV coupling response at fixed masses

The previous calculation varied effective mixed coefficients. This calculation varies the six actual finite-mediator cubic current coefficients while holding the UV source-contact quartics fixed. Each change induces correlated self and cross coefficients in the source EFT. Six independent quadratic coefficients compensate the signed source eigenvalues, preserving the complete finite quark masses and current weights.

The resulting four-by-six observable response has four nonzero singular values. Four independent current directions are varied in both signs, the source stationary and mass constraints are solved again, and the full 48-coordinate Hessian is reconstructed for each mediator action. Every selected solution remains locally positive and preserves the complete quark masses to numerical precision. Finite and implicit responses agree, with the discrepancy norm below the weakest finite singular value. The weakest direction is much smaller than the other three; the receipt retains that numerical separation explicitly.

The algebraic reason is also visible. For the finite-eight and five-pair currents, the two self-anisotropy coefficients and two cross coefficients have parameter Jacobian determinant minus $(a-d-b-c)(a+d+b+c)$. It is generically nonzero. The nine-dimensional channel supplies another independent cross coefficient. The remaining radial self changes can be compensated by allowed radial contacts. Representation safety protects the fermion structure, while those current coefficients still adjust the frame. The Gaussian mechanism therefore does not force the nominated golden and phase constraints.

The ordinary adjoint changes the fermion boundary

The 76-scalar universal completion also includes singlet and ordinary-adjoint mediators. Those representations allow new heavy-mass Yukawa contractions. Setting them to zero reproduces the old protected action, but the old scalar-field charges do not forbid them. The termwise theorem cannot be transferred to the enlarged complete fermion action without checking those vertices.

An exact two-family block example has Hermitian source and heavy-mass vertex matrices, yet its heavy-state paired product is $21/25$ minus $3i/25$. The imaginary part is nonzero despite both individual matrices being Hermitian. At the actual 76-scalar completed vacuum, an equivariant ordinary-adjoint heavy-mass coupling with real coefficient 0.05 gives a neutral-scalar first-order mass phase approximately minus 7.41 times ten to the minus eleven. The complete scalar Hessian and finite Dirac spectrum are used. This is an explicit neutral-scalar threshold, not an all-diagram strong-CP matching calculation or an experimental limit claim.

Thus finite eight-prime, nine and five-pair mediators retain the stated renormalizable protection automatically, while ordinary adjoints introduce a new allowed interaction that the previous theorem did not cover. Singlet heavy-mass couplings require their own analysis; this example does not assert that every additional allowed coupling produces a phase.

Scalar-loop algebra and the enforced SU(3) relation

For canonical real scalars the scalar-only quartic beta function, multiplied by sixteen pi squared, is one half the trace of the squared potential Hessian. Polarizing this contraction defines a commutative, generally nonassociative product on quartic operators. The compiler reconstructs the 52-dimensional source algebra from polynomial Hessians and validates the identities at independent points. Its floating-point reconstruction agrees with rational coefficient candidates to better than ten to the minus ten; the rational table is a numerically validated representation, not an exact symbolic finite-group tensor proof. Single-field and eight-field radial coefficients independently fix the normalization.

Two separate radial quartics form a two-dimensional closed sector. Adding only the finite-eight cross term generates dimensions 3, 4, 8 and finally 9. The closure contains both self-anisotropies, the radial cross product, the ordinary adjoint quadratic-current product and all three finite cross channels. Starting with all three finite channels gives the same closure. In particular, omitting the nine and five-pair cross operators is not a scalar-loop invariant truncation of this finite-eight interaction.

The three finite projectors sum to the SU(3) 27 projector. A single unsplit SU(3) 27 mediator has one quadratic covariant per sector and enforces equal coefficients in the finite-eight, nine and CP-paired five channels. On two traceless Hermitian sources, the corresponding mixed invariant is $\text{Tr}(AB)$ squared minus six fifths $\text{Tr}(A \text{ squared } B \text{ squared})$ plus eleven fortieths $\text{Tr}(A \text{ squared}) \text{Tr}(B \text{ squared})$. Its self contraction is $27/40$ times the squared adjoint norm. The scalar-loop

compiler confirms a five-dimensional closed subalgebra for this unified interaction and the two radial terms.

This is a genuine coefficient relation with a representation origin and a scalar-loop stability mechanism. It is not the golden relation. At fixed nondegenerate spectra the SU(3) trace quartics have moment-space Hessian rank at most one, so they cannot select an isolated interior CP-breaking relative frame. Splitting the 27 under the finite family group permits three independent masses and corresponding coefficient ratios. The 251-coefficient census and actual UV response show precisely which freedoms must then be constrained.

Exact running identity and scope

For two real scalars with quartic coefficients a , b and c multiplying x to the fourth, x squared y squared and y to the fourth, the scalar-only beta coefficients are $72a$ squared + $2b$ squared, $24b(a+c) + 16b$ squared and $72c$ squared + $2b$ squared. On the rank-one surface $a = k u$ squared, $b = 2k u v$ and $c = k v$ squared, the derivative of b squared minus four $a c$ is minus $128k$ cubed u squared v squared $(u-v)$ squared. Generic rank-one exchange identities are therefore not preserved as those same tree relations by light-scalar running. The nontrivial radial case $u=v$ is preserved.

The source model's scalar counterterm direction also changes its golden depth under a fixed-mass compensation. That response is a naturalness diagnostic; it is not a complete CKM renormalization-group evolution. Gauge terms, Yukawa running, Higgs contributions, field normalization and matching thresholds must be included before interpreting a flow as a physical CKM prediction.

Reproduction and references

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_valentiner_mediator_closure.py`. The scientific receipt is `receipts/m22_interactions/valentiner_mediator_closure.json`; the reusable multiplication table is `nonet_scalar_loop_algebra.json` in the same directory. The modules `nonet_mediators.py` and `quartic_renormalization.py` implement constructive matching, exact operator counts, scalar-loop multiplication and generated-subalgebra searches. Focused checks cover current Jacobians, source contacts, positive Schur complements, actual fixed-mass UV deformations, the fermion boundary, exact soft/low-energy counterterms and independent polynomial evaluations.

The scalar-loop algebra viewpoint has prior literature: Nadia Flodgren and Bo Sundborg, One-loop algebras and fixed flow trajectories in adjoint multi-scalar gauge theory, arXiv:2303.13884. RG stability without an evident symmetry is also a real possibility, examined by Howard Haber and P. M. Ferreira, RG-stable parameter relations of a scalar field theory in absence of a symmetry, arXiv:2502.11011. Those works motivate testing closure rather than assuming every unsupported tree relation must fail. The 52-operator compilation, finite-group closure, exact 251-coefficient census and completed-vacuum response here are model-specific calculations; independent novelty review remains separate.